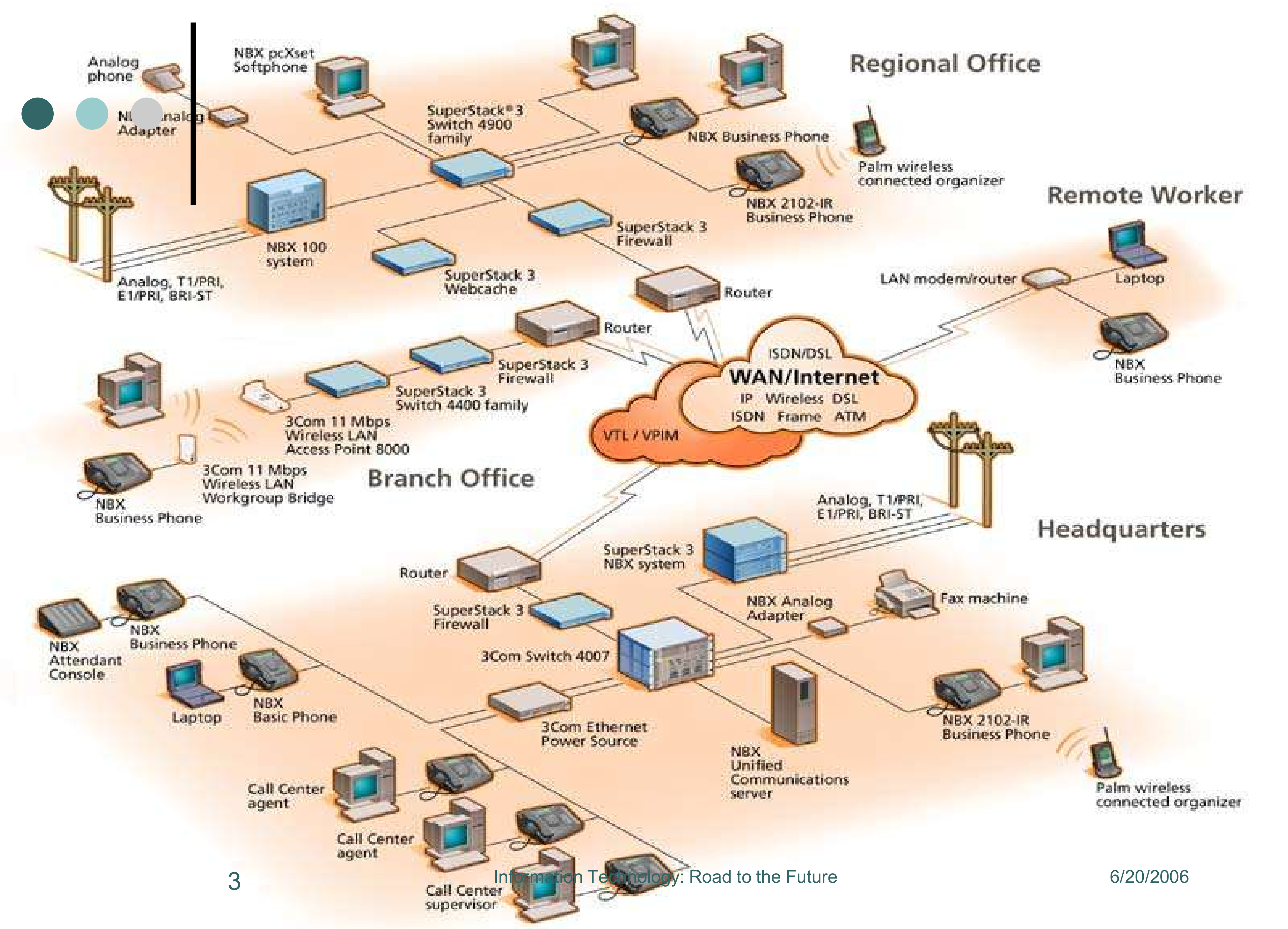
Information Technology: A Road To The Future

Jim Pham

6/20/2006

What is Information Technology

- Physical Infrastructure
  - Client Server Systems
  - Networks
  - Relational Databases
  - Operating Software
  - Application Software
- Technology-based Knowledge


Analog phone
NBX pcXset Softphone
Adapter
SuperStack® 3 Switch 4900 family
Regional Office
NBX Business Phone
Palm wireless connected organizer
NBX 2102-IR Business Phone
Remote Worker
NBX 100 system
SuperStack 3 Firewall
Analog, T1/PRI, E1/PRI, BRI-ST
SuperStack 3 Webcache
Router
LAN modem/router
Laptop
Router
ISDN/DSL
WAN/Internet
IP Wireless DSL
ISDN Frame ATM
VTL / VPIM
SuperStack 3 Firewall
SuperStack 3 Switch 4400 family
NBX Business Phone
3Com 11 Mbps Wireless LAN Access Point 8000
3Com 11 Mbps Wireless LAN Workgroup Bridge
Branch Office
NBX Business Phone
Analog, T1/PRI, E1/PRI, BRI-ST
Headquarters
SuperStack 3 NBX system
Router
NBX Analog Adapter
Fax machine
SuperStack 3 Firewall
NBX Attendant Console
NBX Business Phone
3Com Switch 4007
Laptop
NBX Basic Phone
3Com Ethernet Power Source
NBX Unified Communications server
NBX 2102-IR Business Phone
Palm wireless connected organizer
Call Center agent
Call Center agent
Call Center supervisor

Historical View of the Technology

- Film in the 1920s
- Television in the late 1950s
- Early in the 1970s, calculators replaced slide-rules as the math tool
- Computer in the 1980s
- Information technology 1990s

Transformation in Higher Education

- 1st Revolution: word processing, spreadsheet, digital imaging, desktop publishing, email, multimedia (1980)
- 2nd Revolution: communication gateway—databases, image and text libraries, videos, other information over the computer networks (1990)
- Today's Information Technology—anything and everything are accessible over the Internet

Benefits of Information Technology

- Information-rich environment
- Improve productivity
- Enhance student learning
- Reduce the cost of instruction
- Widespread academic uses of information technology—content, curriculum, and pedagogy

Information Technology Investment

IT IS WORTH IT?

The Campus Environment

The national system of schools, colleges, and universities is undergoing a major transformation. This transformation is inevitable, irreversible, and unpredictable, although we can still influence its direction. The emergence of new information technologies is neither the cause, the purpose, nor the consequence of this transformation. (pg 12)

Technology and the Quest for Productivity

"Institutions must not continue to underestimate the real costs, complexity, and duration of the successful implementation process. Without understanding these cycles and their costs, campuses will neither recognize nor attain the full benefits that technology might offer to students, faculty, curriculum, and institutional effectiveness."

Motivating Behaviors

- Competitive position
- Teaching and learning
- Curriculum enhancement
- Student preparation for the labor market

Faculty Perception

"…faculty members are reluctant to move beyond word-processing because they believe (wrongly) that information technology will not be terribly important for the courses they teach for the next five to ten years."

Change and Expectation

What can you, as a leader, do to ensure you and your institution are on the road to transformation?

Personalize Learning

- A culture of Lifelong Learning
- Access to lifelong learning
- Content to support individual lifelong learners
- A social context for lifelong learning
- Distance learning

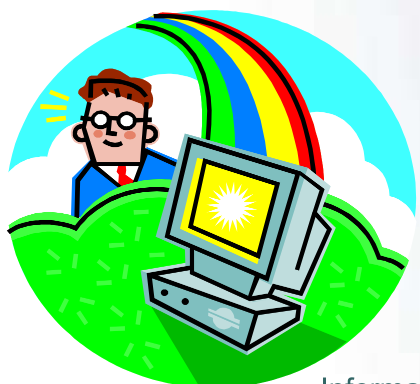
Where is Technology Headed?

- Colleges and universities have some form of connection to the Internet
- Portable computer and related mobile devices will continue to stimulate teaching and learning
- Students and faculty have email access
- Information technology is available instantly at your fingertips
- Information technology enhances the learning process through collaborative/cooperative learning
- Teachers as Education coaches--presenters and mentors

Conclusion

I agree with this review, which suggests that content, curriculum, and communication—rather than productivity—are the appropriate focus of and rationale for campus investments in information technology. In addition, information technology continues to shape the colleges and universities; therefore, it is important for future leaders in education to form partnerships with the industry.

"In the future learning will not be limited by the walls of the classroom, the hours of the day or the confines of the students home … it will happen Anytime Anywhere"



Bill Gates - Microsoft